



香港中文大學(深圳)  
The Chinese University of Hong Kong, Shenzhen

## **DDA2001: Introduction to Data Science**

# **Introduction**

**2025/01/07**

**Shuang Li**

# What is Data Science?

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- Let's make an analogy 😊



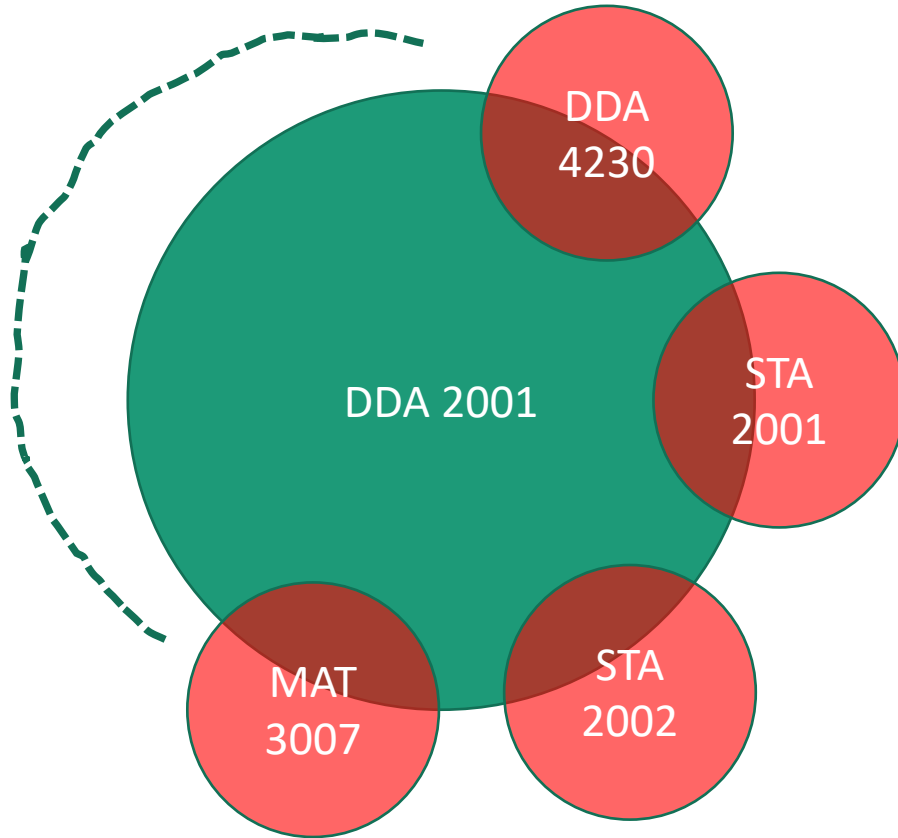
# What is Data Science?

- Similar key steps 😊
- **Data Collection - Sourcing Quality Ingredients**
- **Data Cleaning - Sorting and Peeling**
- **Data Preparation - Chopping and Slicing**
- **Train Model - Cooking**
- **Test Data - Guest Taste Testing**



A chef in a white uniform is sprinkling black pepper from a wooden mill into a white bowl containing a creamy substance. The background is a white subway tile wall. Overlaid on the image is a word cloud of machine learning terms, with "deep Learning" and "svm" being the most prominent.

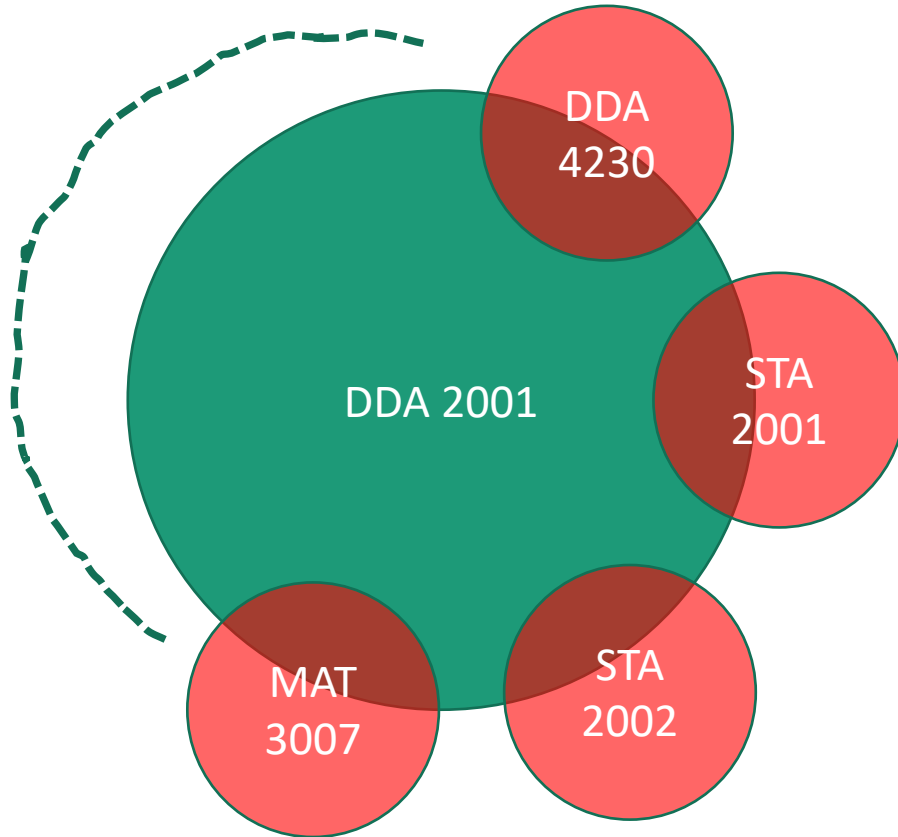
- DDA 2001 is an introductory course for advanced courses.



There are overlaps and the overlaps just include

- Key concepts associated with **simple** mathematical calculations
- Advanced concepts that require **fewer** or **no** mathematical calculations

- DDA 2001 is an introductory course for advanced courses.



- You will participate in exercises to help you understand the key concepts and apply them in real-world situations.
- We'll provide examples to make complex ideas clearer. If you're interested, you can dive deeper into these topics later on.

# Why do we need to take DDA 2001?

When you focus too much on the details..



But forget that the full picture is important!  
Sometimes, you need a bird's-eye view to see how all the pieces fit together.

# After taking this course

- You'll get the big picture of what data science is all about.
- We'll break down its different areas so you can see what makes each one tick.
- By the end, you'll know which area sparks your interest, helping you choose a major that truly aligns with your passions!





# An appetizer of data science application

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The Math Behind Basketball's Wildest  
Moves | Rajiv Maheswaran | TED Talks

[https://www.youtube.com/watch?v=66ko\\_cWSHBU&ab\\_channel=TED](https://www.youtube.com/watch?v=66ko_cWSHBU&ab_channel=TED) (6:40-8:25)



# From the video, you may think by yourself:

- How can data help coaches make better decisions during a game?
- What kind of data about players could be useful to understand their performance?
- How might analyzing player statistics help improve team strategy?
- What challenges do you think analysts might face when using data to predict game outcomes or player performance?
- Do you think data can replace a coach's experience and gut feeling? Why or why not?
- Why is it important to collect data over multiple games or seasons, instead of just one?
- ...



# Course Organization

# Course Staff (Lecturer)

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- Instructor: Shuang Li
  - Office: Daoyuan Building 508B
  - Office Hours: Friday 3:00 pm – 4.00 pm
  - Email: [lishuang@cuhk.edu.cn](mailto:lishuang@cuhk.edu.cn)
- Lectures:
  - Room: Teaching Complex A401
  - Times: Tuesday, Thursday 13:30 – 14:50
-

# Course staff (TAs)

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Behind  
every teacher  
IS A GREAT  
TEACHING  
ASSISTANT



|                   |                            |
|-------------------|----------------------------|
| • Sheng Xu        | 徐圣                         |
| Email:            | 223040246@link.cuhk.edu.cn |
| • Bo Yue          | 岳博                         |
| Email:            | 224040353@link.cuhk.edu.cn |
| • Jiaming Huang   | 黄嘉铭                        |
| Email:            | 224040352@link.cuhk.edu.cn |
| • Ruixing Zhuzhou | 朱周睿星                       |
| Email:            | 222010522@link.cuhk.edu.cn |
| • Yuwei Xu        | 徐昱玮                        |
| Email:            | 224040360@link.cuhk.edu.cn |
| • Fei Yu          | 余菲                         |
| Email:            | 222043013@link.cuhk.edu.cn |
| • Shu Wang        | 王澍                         |
| Email:            | 223040238@link.cuhk.edu.cn |
| • Yiheng Wang     | 王一衡                        |
| Email:            | 224040341@link.cuhk.edu.cn |

# Schedule

| Time          | Monday                | Tuesday               | Wednesday             | Thursday              | Friday |
|---------------|-----------------------|-----------------------|-----------------------|-----------------------|--------|
| 8:30 – 9:00   |                       |                       |                       |                       |        |
| 9:00 – 9:30   |                       |                       |                       |                       |        |
| 9:30 – 10:00  |                       |                       |                       |                       |        |
| 10:00 – 10:30 |                       |                       |                       |                       |        |
| 10:30 – 11:00 | Lecture 04            | Lecture 02            | Lecture 04            | Lecture 02            |        |
| 11:00 – 11:30 | Jiaqi Lu              | Pin Gao               | Jiaqi Lu              | Pin Gao               |        |
| 11:30 – 11:50 | Teaching Complex A401 | Teaching Complex A401 | Teaching Complex A401 | Teaching Complex A401 |        |
| 12:00 – 12:30 |                       |                       |                       |                       |        |
| 12:30 – 13:00 |                       |                       |                       |                       |        |
| 13:00 – 13:30 |                       |                       |                       |                       |        |
| 13:30 – 14:00 | Lecture 01            | Lecture 03            | Lecture 01            | Lecture 03            |        |
| 14:00 – 14:30 | Zicheng Wang          | Shuang Li             | Zicheng Wang          | Shuang Li             |        |
| 14:30 – 14:50 | Teaching Complex A401 | Teaching Complex A401 | Teaching Complex A401 | Teaching Complex A401 |        |
| 15:00 – 15:30 |                       |                       |                       |                       |        |
| 15:30 – 16:00 |                       |                       |                       |                       |        |
| 16:00 – 16:30 |                       |                       |                       |                       |        |
| 16:30 – 17:00 |                       |                       |                       |                       |        |
| 17:00 – 17:30 |                       |                       |                       |                       |        |
| 18:00 – 18:30 |                       | Tutorial 03           |                       |                       |        |
| 18:30 – 18:50 |                       | Teaching B 105        |                       |                       |        |
| 19:00 – 19:30 | Tutorial 01           | Tutorial 04           | Tutorial 05           | Tutorial 07           |        |
| 19:30 – 19:50 | Teaching B 105        | Teaching B 105        | Teaching B 105        | Teaching B 105        |        |
| 20:00 – 20:30 | Tutorial 02           |                       | Tutorial 06           | Tutorial 08           |        |
| 20:30 – 20:50 | Teaching B 105        |                       | Teaching B 105        | Teaching B 105        |        |

# Tutorials

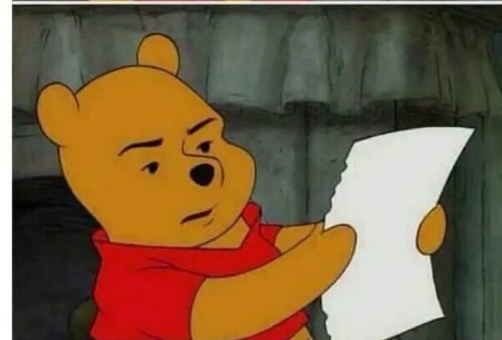
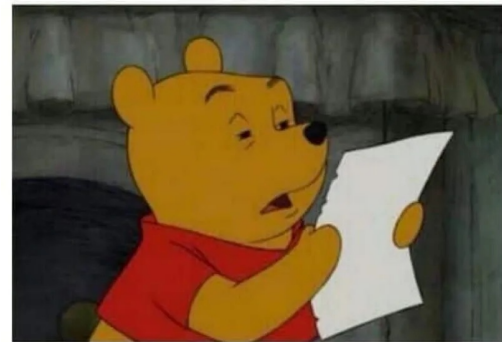
- Tutorials start on the **second** week

|              |                  |                         |          |                   |
|--------------|------------------|-------------------------|----------|-------------------|
| Probability  | Tutorial Session | TA                      | Location | Time              |
|              | T01              | Shu Wang, Yuwei Xu      | TB 105   | Mon: 19:00–19:50  |
|              | T02              | Shu Wang, Yuwei Xu      | TB 105   | Mon: 20:00–20:50  |
|              | T03              | Shu Wang, Yuwei Xu      | TB 105   | Tues: 18:00–18:50 |
|              | T04              | Shu Wang, Yuwei Xu      | TB 105   | Tues: 19:00–19:50 |
|              | T05              | Shu Wang, Yuwei Xu      | TB 105   | Wed: 19:00–19:50  |
|              | T06              | Shu Wang, Yuwei Xu      | TB 105   | Wed: 20:00–20:50  |
|              | T07              | Shu Wang, Yuwei Xu      | TB 105   | Thu: 19:00–19:50  |
| Statistics   | Tutorial Session | TA                      | Location | Time              |
|              | T01              | Sheng Xu, Jiaming Huang | TB 105   | Mon: 19:00–19:50  |
|              | T02              | Sheng Xu, Jiaming Huang | TB 105   | Mon: 20:00–20:50  |
|              | T03              | Sheng Xu, Jiaming Huang | TB 105   | Tues: 18:00–18:50 |
|              | T04              | Sheng Xu, Jiaming Huang | TB 105   | Tues: 19:00–19:50 |
|              | T05              | Sheng Xu, Jiaming Huang | TB 105   | Wed: 19:00–19:50  |
|              | T06              | Sheng Xu, Jiaming Huang | TB 105   | Wed: 20:00–20:50  |
|              | T07              | Sheng Xu, Jiaming Huang | TB 105   | Thu: 19:00–19:50  |
| Optimization | Tutorial Session | TA                      | Location | Time              |
|              | T01              | Yiheng Wang, Bo Yue     | TB 105   | Mon: 19:00–19:50  |
|              | T02              | Yiheng Wang, Bo Yue     | TB 105   | Mon: 20:00–20:50  |
|              | T03              | Yiheng Wang, Bo Yue     | TB 105   | Tues: 18:00–18:50 |
|              | T04              | Yiheng Wang, Bo Yue     | TB 105   | Tues: 19:00–19:50 |
|              | T05              | Yiheng Wang, Bo Yue     | TB 105   | Wed: 19:00–19:50  |
|              | T06              | Yiheng Wang, Bo Yue     | TB 105   | Wed: 20:00–20:50  |
|              | T07              | Yiheng Wang, Bo Yue     | TB 105   | Thu: 19:00–19:50  |
| ML           | Tutorial Session | TA                      | Location | Time              |
|              | T01              | Zhouruixing Zhu, Yu Fei | TB 105   | Mon: 19:00–19:50  |
|              | T02              | Zhouruixing Zhu, Yu Fei | TB 105   | Mon: 20:00–20:50  |
|              | T03              | Zhouruixing Zhu, Yu Fei | TB 105   | Tues: 18:00–18:50 |
|              | T04              | Zhouruixing Zhu, Yu Fei | TB 105   | Tues: 19:00–19:50 |
|              | T05              | Zhouruixing Zhu, Yu Fei | TB 105   | Wed: 19:00–19:50  |
|              | T06              | Zhouruixing Zhu, Yu Fei | TB 105   | Wed: 20:00–20:50  |
|              | T07              | Zhouruixing Zhu, Yu Fei | TB 105   | Thu: 19:00–19:50  |
|              | T08              | Zhouruixing Zhu, Yu Fei | TB 105   | Thu: 20:00–20:50  |

# Tentative Calendar

| Weeks | Content/ topic/ activity                                                                                                                                                                                                                                                                                                                                              |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Course Introduction</b>                                                                                                                                                                                                                                                                                                                                            |
| 1     | Course plan, grading, overview, examples, and applications, etc.                                                                                                                                                                                                                                                                                                      |
|       | <b>Probability</b>                                                                                                                                                                                                                                                                                                                                                    |
| 3     | 1 Terminologies: sample spaces, probability function, event, etc..<br>2. Main Properties: mean, variance, CDF.<br>3. Common Discrete RV: Bernoulli, Binomial, Poisson, Geometric.<br>4. Common Continuous RV: uniform, normal, exponential.<br>5*. Advanced Concepts: correlation and conditional probability (Bayes rule).<br>6#. Additional examples and exercises. |
|       | <b>Statistics</b>                                                                                                                                                                                                                                                                                                                                                     |
| 3     | 1. Sampling: Monte Carlo (integration, MDP, etc.).<br>2. Sampling: generating uncommon RV.<br>3. Point estimator: MLE.<br>4. MLE Application: Regression Analysis.<br>5*. Confidence interval<br>6#. Additional examples and exercises.                                                                                                                               |
| 0.5   | <b>Midterm Review</b>                                                                                                                                                                                                                                                                                                                                                 |

Looking at that first syllabus like

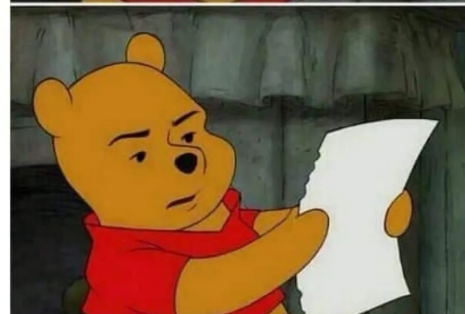
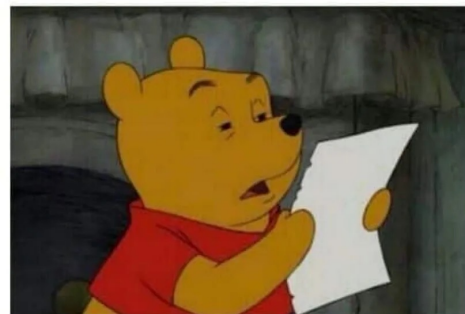




# Tentative Calendar

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Optimization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3.5 | <ol style="list-style-type: none"> <li>1. Problem formulation (objective function, constraints, decision variables, etc.)<br/>Application examples in revenue management, online advertising, etc.</li> <li>2. Optimization: convex set.</li> <li>3. Convex Optimization: convex function.</li> <li>4. Convex Optimization: convex function Property.</li> <li>5. Convex Optimization: quasi-convex function.</li> <li>6*. Gradient Descent and Stochastic Gradient Descent</li> <li>7#. Additional examples and exercises.</li> </ol> |
|     | <b>Machine Learning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.5 | <ol style="list-style-type: none"> <li>1*. Supervised learning: K-Nearest Neighbors (K-NN).</li> <li>2*. Supervised learning: Logistic regression, Neural Nets.</li> <li>3*. Unsupervised learning: Clustering (K means).</li> <li>4**. Cross Validation.</li> <li>5**: Deep Learning and Reinforcement Learning.</li> </ol>                                                                                                                                                                                                           |
| 0.5 | <b>Final Review</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Looking at that first syllabus like



# Problem exercise.

\* Only concepts will be examined.

\*\* Contents that will be covered if time permits. If covered, only concepts will be examined.

# Learning Assessment Strategy

| Component/ method | % weight |
|-------------------|----------|
| Assignments       | 15%      |
| Quizzes           | 15%      |
| Mid-Term Exam     | 30%      |
| Final Exam        | 40%      |

- **3 to 5 Guest Lectures on Fridays (Dates TBD). Quizzes on Guest Lectures.**
- **No late homework will be accepted.**
- **Midterm (tentative): TBD**
- **There will be no make-up midterm exams**
- **Zero tolerance policy when it comes to cheating so, DON'T CHEAT**
- **Missing the midterm or final exam without prior notification to and approval of the instructors will automatically result in the "0" grade for the exam.**



About me

# Shuang Li

- **My academic pathway**

- PhD in Industrial Engineering, Georgia Tech
- Joined SDS of CUHK(SZ) in July 2021, and is currently an Assistant Professor

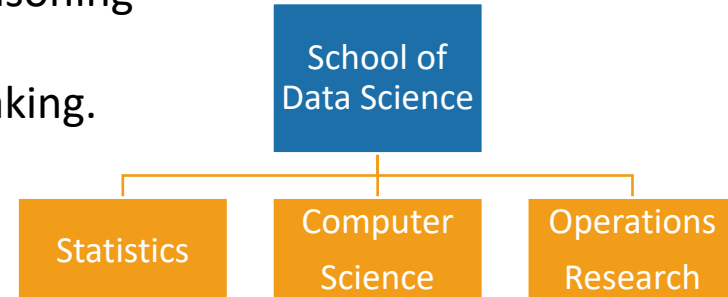
- **My Research Interest related to “Data Science”: Machine Learning**

- **Interpretable ML Models:**

- Enable machine learning to perform logical reasoning

- **Healthcare Applications:**

- Using ML tools to assist doctors in decision-making.



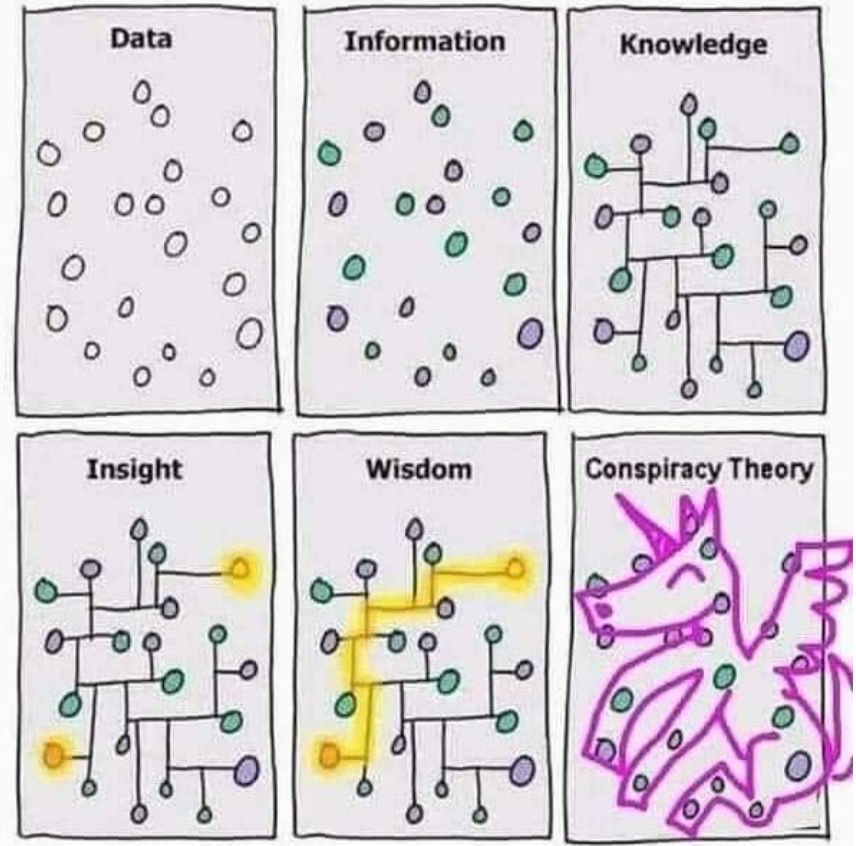
What is Data Science?

数据科学

# What is data science?

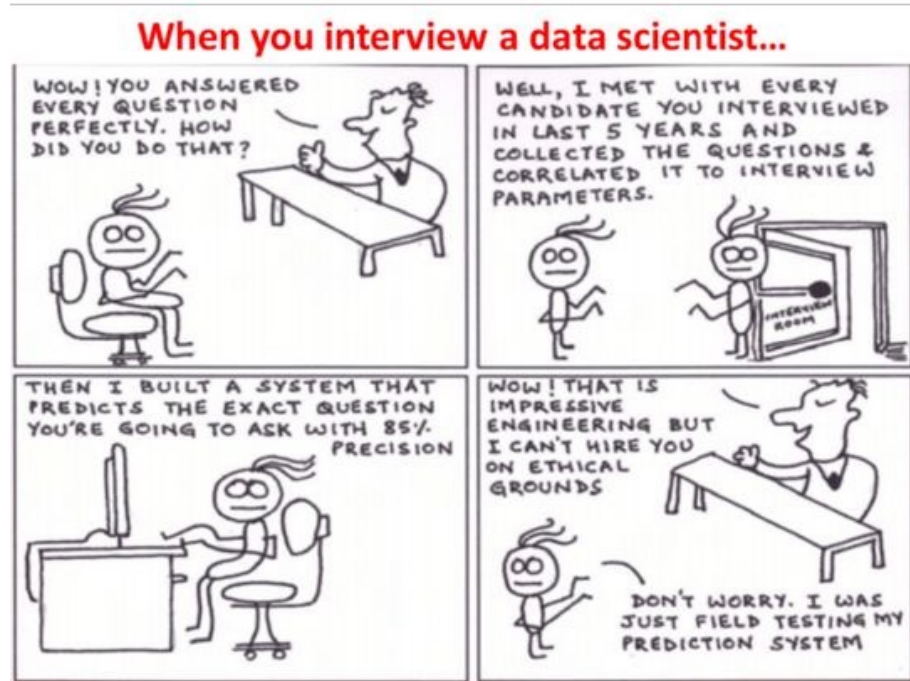
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- Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to **find unseen patterns, derive meaningful information, and make business decisions.**



# What is data science?

- Data science is not one single subject! It incorporates probability, statistics, operations research, machine learning, and business applications...
- You need a solid mathematical/technical foundation, together with some industrial expertise in order to be a data scientist.
- Analytical skills + business understanding.



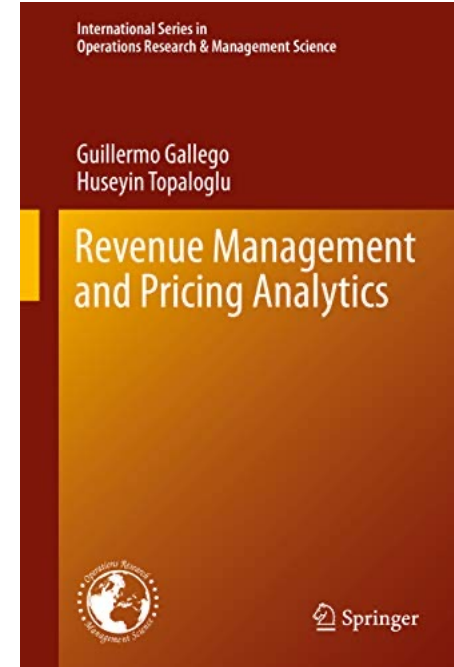
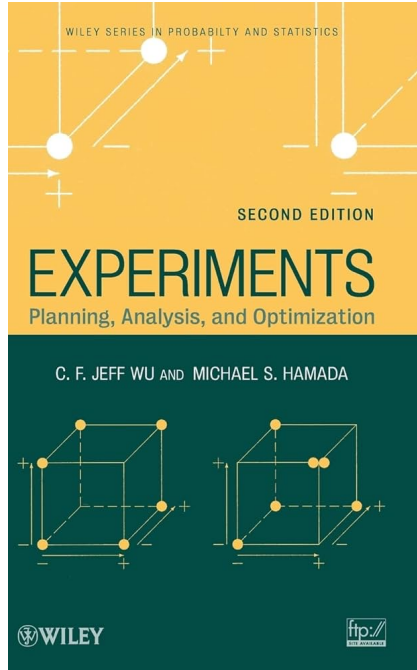
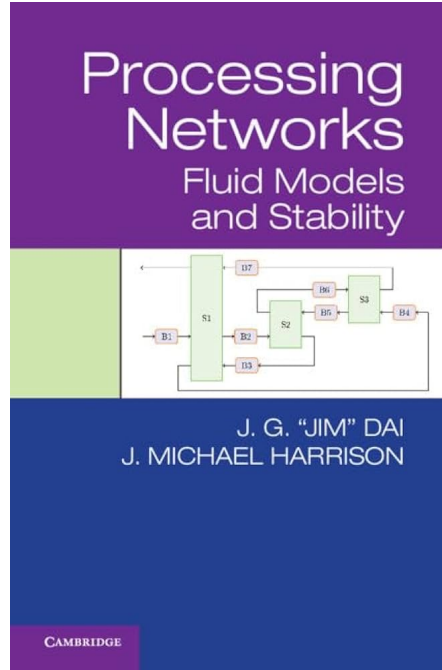
# What is data science: tree pillars

- **Statistics:** The science of collecting, analyzing, and interpreting data to draw meaningful conclusions and predict future trends.
- **Computer Science:** The study of algorithms, data structures, and programming to efficiently process data and solve complex problems.
- **Optimization:** A mathematical approach to finding the best solutions for decision-making problems, improving outcomes like cost, performance, or resource allocation.





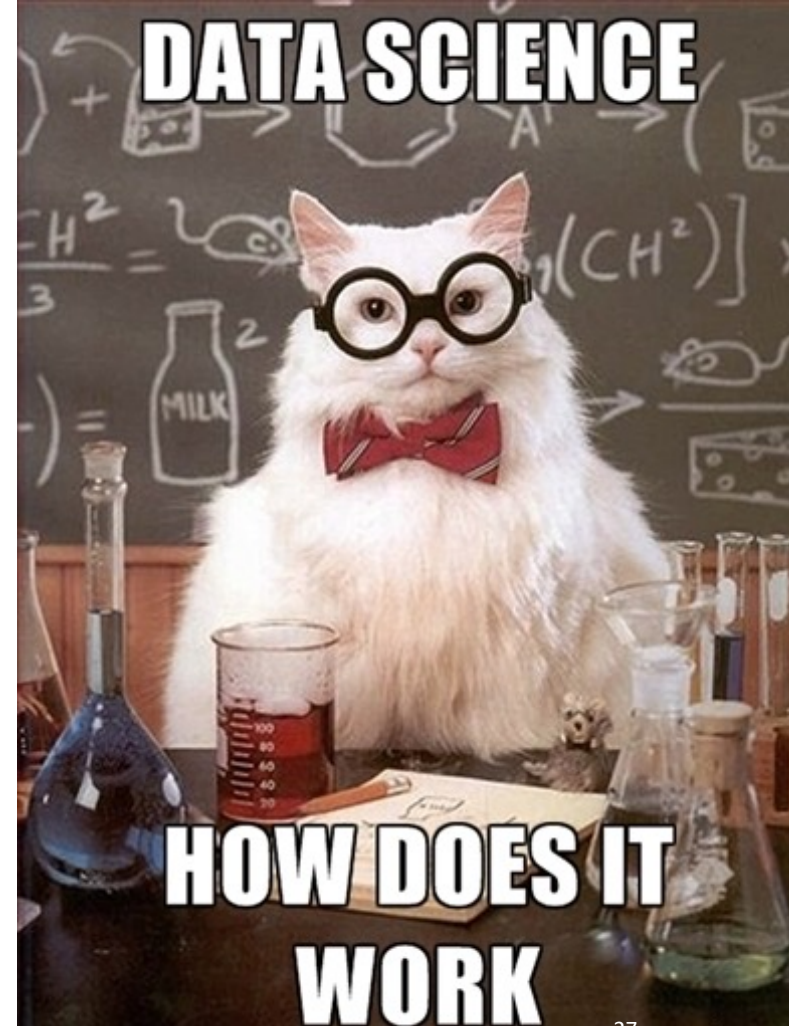
# References





# Course Introduction

Next, you'll start thinking and solving problems like a data scientist!





The festival has just passed.  
Did you buy a lot of things?

Suppose you  
are a merchant



During special sale events like the Global Shopping Festival, when people went crazy buying things. The merchant had no idea how to **price his products to maximize profits** while keeping customers happy. **He needed a plan!**

# Things to consider when choosing a pricing strategy

Cost of Products



Customers



Market Positioning



Competitors



Profit Margins



Geography



Product Offerings



To stand out from the event,



E-commerce merchant

被迫成为了“数学家”

Forced to become mathematicians

Do I have to set the prices for all these?

- Even if you are already good at these pricing techniques, it is still very difficult to implement them.





Has the price not changed?

1、2、3 上链接  
1元3瓶精酿啤酒

1, 2, 3, click the link,  
one yuan for three  
bottles of craft beer

价格没变?

对不起!

Sorry!



Sorry, setting this price was too troublesome. Full refunds for all purchases and free giveaways! Let's try another round!

对不起，这个价格设置太麻烦了  
已购买全部退款免费送！再抢一轮

啥？还是原价，  
又翻车了？

What? Still the original price?  
Crashed again?

对不起，  
我是新人主播

Sorry, I am a new broadcaster.



**Background: SKU (Stock Keeping Unit):** A unique identifier for managing individual products in inventory. It represents specific product attributes (e.g., brand, size, color). SKUs help businesses distinguish product variations, set prices, and track inventory efficiently. For example, each size and color of a shirt would have a unique SKU, possibly with different prices based on demand or cost.

There may be hundreds of SKUs in one live broadcast

For e-commerce platforms, what they are facing can be more than millions of SKUs

Say NO to manual pricing!

所以



**NO!**  
**NO!**

To solve this, the merchant turned to his customers...

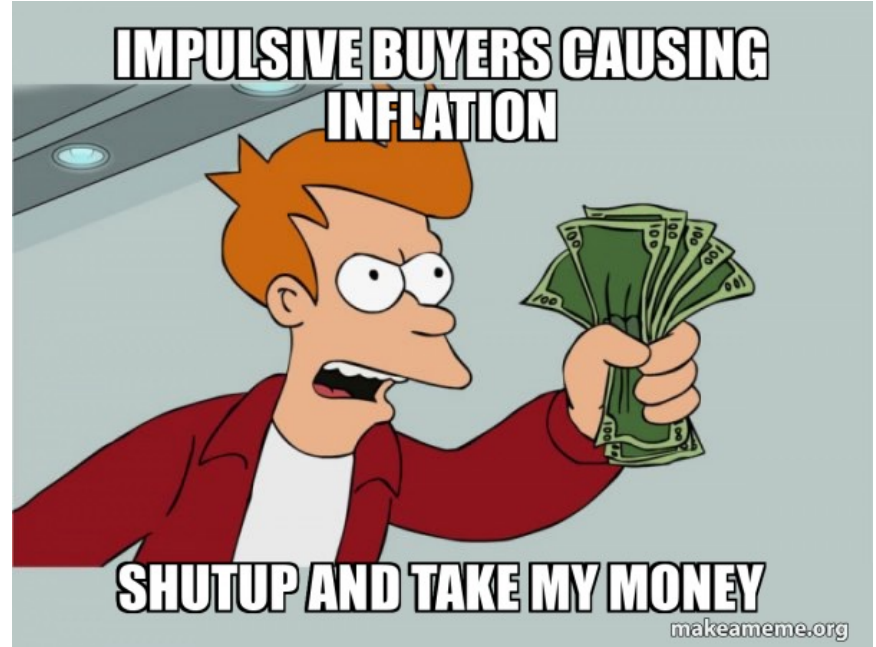
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**The Hand-Chopping People (剁手人 - Duò shǒu rén)**

These were the impulsive buyers, the ones who couldn't resist a good deal. Every time a sale popped up, they'd "chop off their hands" and buy without thinking twice. The merchant knew that these customers loved a bargain, but could easily regret their purchase later.

**The Dirt-Eating People (吃土人 - Chī tǔ rén)**

These customers were a bit like the Hand-Chopping People but with a twist. After spending all their money on shopping, they had nothing left for essentials. This left them metaphorically eating dirt!



# To solve this, the merchant turned to his customers...

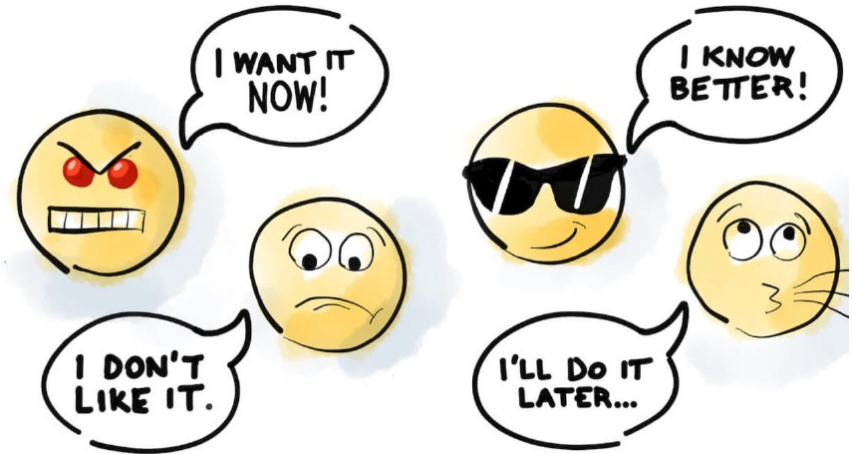
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## The Tail-Payment People (尾款人 - Wěi kuǎn rén)

This translates to "tail payment people". It's used to describe people who make a down payment for a product (often during a sale like Singles' Day in China) and are waiting to pay the balance amount.

## The Late-Eight People (晚八人 - Wǎn bā rén)

This phrase is a bit more context-dependent. Literally, it means "late-eight people". This could refer to people who work late (until 8 PM), or it could be a specific cultural reference that needs more context to accurately translate.



How to make decisions more  
**scientifically and efficiently?**



1. Let a machine think like you





2. Let the machine make decisions like you

# The Challenge: Understanding the Customers

The merchant had to figure out how to set prices that would appeal to each group without overcharging or underpricing them.

That's when he turned to a model: the **Multinomial Logit Model**—a powerful tool in data science.

Given the Multinomial Logit Model, the merchant can

- Understand different consumer types' purchasing preferences.
- Learn which products to recommend to each consumer type.

*(Note that the goal here is not to teach you the multinomial logit model, but to treat it as a data science tool. The focus is on showing how, assuming you are familiar with this tool, you can use it to solve the pricing problem—one of many real-life data science challenges.)*

# Problem Solving: Step 1 - Data Collection

- **Key Question:**

How do you start when solving a problem like pricing strategy?

- **Data Collection**

To make good decisions, you need data. Start by gathering information, such as:

- **Historical Data:** Past purchase behaviors, sales data, and customer preferences.
- **Competitor's Offers:** What are your competitors pricing their products at? What promotions are they running?

- **Why it's Important:**

Data helps us understand customer behavior and market trends, allowing you to design an effective pricing strategy.

## Data collection? **Programming.**

- Obtain historical data, obtain current competitor's offers, etc.

### Key Question:

How do you process and use all this data?

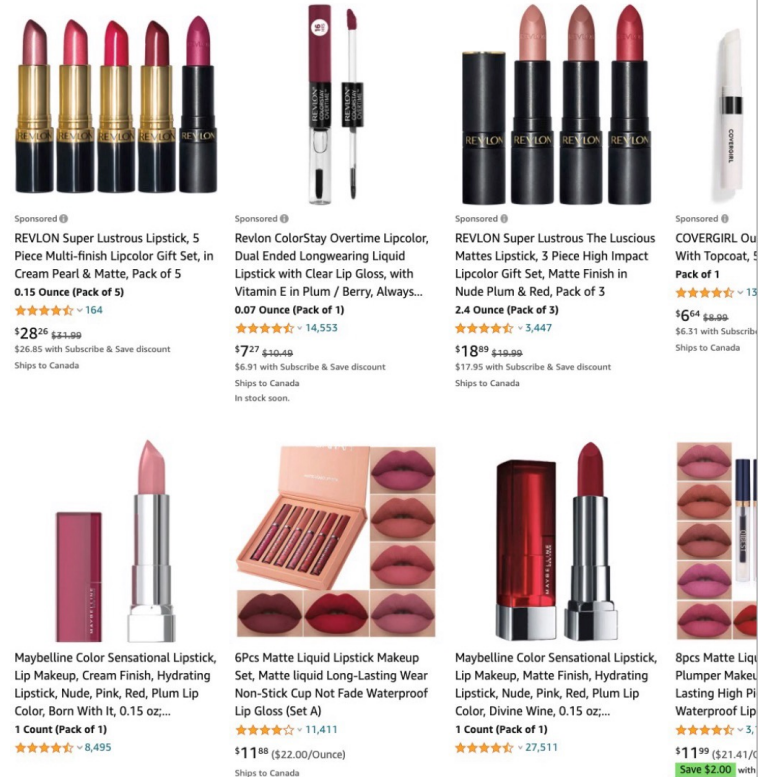
### Programming

Programming involves creating a set of instructions that tell a computer how to perform tasks. Learning how to program allows you to:

- **Automate tasks:** No more manual work—computers can handle repetitive tasks like data analysis.
- **Analyze large datasets:** Work with large volumes of data efficiently.

### Why it's Important:

With programming skills, you can manipulate, clean, and analyze data more quickly and effectively.



**Revlon Super Lustrous Lipstick, 5 Piece Multi-finish Lipcolor Gift Set, in Cream Pearl & Matte, Pack of 5**  
0.15 Ounce (Pack of 5)  
\$28.26 ~~\$31.99~~  
\$26.85 with Subscribe & Save discount  
Ships to Canada  
★★★★☆ ~ 164

**Revlon ColorStay Overtime Lipcolor, Dual Ended Longwearing Liquid Lipstick with Clear Lip Gloss, with Vitamin E in Plum / Berry, Always...**  
0.07 Ounce (Pack of 1)  
\$7.27 ~~\$7.49~~  
\$6.91 with Subscribe & Save discount  
In stock soon.  
★★★★☆ ~ 14,553

**Revlon Super Lustrous The Luscious Mattes Lipstick, 3 Piece High Impact Lipcolor Gift Set, Matte Finish in Nude Plum & Red, Pack of 3**  
2.4 Ounce (Pack of 3)  
\$18.89 ~~\$19.99~~  
\$17.95 with Subscribe & Save discount  
Ships to Canada  
★★★★☆ ~ 3,447

**Covergirl Out With Topcoat, 5 Pack of 1**  
\$6.64 ~~\$8.99~~  
\$6.31 with Subscribe & Save discount  
Ships to Canada  
★★★★☆ ~ 13

**Maybelline Color Sensational Lipstick, Lip Makeup, Cream Finish, Hydrating Lipstick, Nude, Pink, Red, Plum Lip Color, Born With It, 0.15 oz;...**  
1 Count (Pack of 1)  
\$11.88 ~~\$12.00~~  
Ships to Canada  
★★★★☆ ~ 8,495

**6Pcs Matte Liquid Lipstick Makeup Set, Matte liquid Long-Lasting Wear Non-Stick Cup Not Fade Waterproof Lip Gloss (Set A)**  
\$11.88 ~~\$22.00~~  
Ships to Canada  
★★★★☆ ~ 11,411

**Maybelline Color Sensational Lipstick, Lip Makeup, Matte Finish, Hydrating Lipstick, Nude, Pink, Red, Plum Lip Color, Divine Wine, 0.15 oz;...**  
1 Count (Pack of 1)  
\$11.99 ~~\$21.41~~  
Save \$2.00 with  
★★★★☆ ~ 27,511

**8pcs Matte Liquid Lipstick Makeup Set, Matte liquid Long-Lasting Wear Non-Stick Cup Not Fade Waterproof Lip Gloss (Set B)**  
\$11.99 ~~\$21.41~~  
Save \$2.00 with  
★★★★☆ ~ 3,447

# Problem Solving: Step 2 - Describing Data Patterns (Probability Theory)

- **Key Question:**

How do we describe and understand the data?

- **Probability Theory**

To understand customer behavior, we need a structured way to **describe uncertainty**. Probability allows us to model the **likelihood** of different events, such as:

- **How do customers choose from a set of products?**
- **What factors influence their purchasing decision?**

- Example: The **Multinomial Logit Model (MNL)** helps describe how customers choose products. The choice probability is:

$$P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$$

where:

- $a_i$  represents the attractiveness/utility/preference score of product  $i$
- $S$  is the set of all products or a product set offered
- The likelihood of choosing a product is proportional to its utility

- MNL is simple but useful. It is widely used and recognized in real life.

**Daniel Little McFadden** (born July 29, 1937) is an American [econometrician](#) who shared the 2000 [Nobel Memorial Prize in Economic Sciences](#) with [James Heckman](#). McFadden's share of the prize was "for his development of theory and methods for analyzing [discrete choice](#)"



# Problem Solving: Choosing a Reliable Data Pattern (Statistics)

- **Key Question:**

How do we know if the data pattern is reliable?

- **Statistics**

Once you've defined a pattern, statistics can help you assess its quality and reliability. You can use statistical tests to:

- **Estimate parameters:** Determine the values of parameters like  $a_i$  in the multinomial logit model.
- **Validate the model:** Ensure that the probability model actually reflects real-world consumer behavior.

- **Why it's Important:**

Statistics helps you test your assumptions and improve the accuracy of your model.



# Problem Solving: Learning a Complicated Model (Machine Learning)

- **Key Question:**  
What if the relationship between product choice and consumer behavior is more complex?
- **Machine Learning**  
If the simple model (like the multinomial logit model) isn't enough, machine learning can help.
- For example:
- $P_i(S)$  may depend on more complex features like:
  - Product attributes (e.g., color, price, brand)
  - Consumer profiles (e.g., age, location, past behavior)
- Machine learning can learn these complex relationships from the data and make better predictions.
- **Why it's Important:**  
Machine learning enables us to capture more complex patterns and make more accurate predictions about customer behavior.

# Problem Solving: Optimization

- **Key Question:**

Now that we understand customer preferences, how do we make the best decision?

- **Optimization**

After building a model and predicting how customers will behave, you want to **maximize** your objectives—such as maximizing revenue or sales.

- For example, you want to choose the products to offer that will maximize your revenue:

$$\max_S \sum_{i \in S} r_i P_i(S)$$

where:

- Suppose the choice rule  $P_i(S)$  is given.
- Let  $r_i$  be the revenue of product  $i$ .
- Choose a set of products to maximize the above objective.

- **Why it's Important:**

Optimization helps you make decisions that achieve the best possible outcome, such as maximizing sales or profits.

# Summary: Solving Problems with Data Science

1. Data collection? **Programming.**
2. How to describe the data pattern? Probability theory.
3. How to choose a reliable data pattern? Statistics.
4. How to learn a complicated function? Machine Learning.
5. What decision should be made to maximize the objective?  
Optimization

Data science is a powerful toolkit for solving problems, and with each step, you gain a deeper understanding of how to make better decisions based on data.

# Data science

- Let a machine think like you
  - Extract knowledge from **data** sets
- Let the machine make decisions like you
  - Apply the knowledge and actionable insights from data to solve problems in a wide range of application domains

# To be a good data scientist,

- Programming languages + analytical skills + business understanding
- It incorporates probability, statistics, operations research, machine learning, and business applications...



- Programming is a fundamental part of data science.
- Programming Languages for Data Science: python, R, C, C++, ...
- In this course,
  - We will give some coding demos
  - There will be **no** coding questions in the homework and the exams.

Collect past  
data

Programming: automate decision.

Propose some  
models

Probability: quantify uncertainty.

Choose the  
best model

Statistics: test credibility.

Prediction for  
given input

Sampling: calculate complicated objective.

Optimize  
input

Optimization: optimize objectives.

Make a decision

Collect past  
data

Programming: automate decision.

No models to propose?

Propose some  
models

Probability: quantify uncertainty.

Choose the  
best model

Statistics: test credibility.

Prediction for  
given input

Sampling: calculate complicated objective.

Optimize  
input

Optimization: optimize objectives.

Make a decision



Collect past  
data

Programming: automate decision.

Propose some  
black boxes

Neural network  
Support vector machine

Optimize the  
parameters

Deep learning

Prediction for  
given input

Sampling: calculate complicated objective.

Optimize  
input

Optimization: optimize objectives.

Make a decision

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